BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Data Analysis Report

Project ID: IT22339010

Project Title: Personalized Food Expiry Prediction

Introduction

1.1. Background:

Household food waste and food safety risks are significantly influenced by the use of static printed expiry dates, which do not adapt to real-world storage conditions or individual food handling behaviors. Printed expiry labels assume ideal storage environments and uniform usage patterns, often resulting in premature food disposal or unsafe consumption.

The Personalized Food Expiry Predictor is designed to address these limitations by analyzing food storage data, environmental context, and safety reference knowledge to generate adaptive yet safe expiry predictions. This report presents the data analysis conducted to support the development of this component, focusing on dataset selection, preprocessing, feature extraction, exploratory analysis, and interpretation of results.

2.

2.1. Research Problem:

Static expiry dates fail to reflect variations in storage conditions, food categories, and user behavior. This leads to inaccurate shelf-life estimation, increased food waste, and potential health risks.

2.2. Objectives:

Objective	Objective number
Analyze food storage and spoilage patterns	1
Predict food expiry duration	2
Validate expiry using safety constraints	3
Adapt expiry predictions using user feedback	4
Prioritize food items for consumption	5

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3. Data Exploration
   3.1. Data Collection

Two datasets were used for this analysis:

- **Food Expiry Predictor Dataset**
  Collected from a publicly available Kaggle dataset, this dataset contains historical food records with attributes related to food type, category, storage condition, and expiry behavior. It serves as the primary dataset for identifying spoilage patterns and supporting expiry prediction.

- **Item Base Expiry Dataset**
  Derived from food safety reference documents and domain literature, this dataset defines biologically safe shelf-life limits for food items under different storage conditions (pantry, refrigerator, freezer). It serves as a safety reference rather than a predictive dataset.

3.2. Dataset Description:

Data Source	Description	Resource	Size	Key Attributes
Food Expiry Dataset	ML training dataset for expiry prediction	Kaggle Dataset	2,102 records	item_name, category, storage, env_features, days_until_expiry
Item Base Expiry Dataset	Biological shelf-life limits	Food Safety Guidelines & Shelf-Life Standards	653 records	base_fridge_days, base_freezer_days, base_pantry_days

3.3. Suitability Analysis

3.3.1. Relevance to Individual Research Objectives:

All datasets align with project objectives:

Objective	Food Expiry Dataset	Item Based Expiry Dataset
Analyze food storage and spoilage patterns	✓	
Predict food expiry duration	✓	✓
Validate expiry using safety constraints		✓
Adapt expiry predictions using user feedback	✓	✓
Prioritize food items for consumption	✓	

4. Methodology

4.1. Data Preprocessing:

Transformation Technique	Food Expiry Dataset	Item Base Dataset
Data Cleaning	✓	✓
Handling Missing Data	✓	
Data Type Conversion	✓	✓
Data Normalization	✓	
Feature Engineering	✓	
Data Encoding	✓	
Data Aggregation	✓	

The original Food Expiry Predictor dataset contained 2,102 records. After removing duplicate entries, inconsistent expiry values, and records with incomplete storage metadata, the final usable dataset consisted of 2,048 records used for modelling and evaluation. This cleaning process ensured data integrity while maintaining sufficient sample size for robust machine learning training.

A total of 54 records (approximately 2.57% of the original dataset) were removed during preprocessing due to duplication, inconsistent expiry values, or incomplete storage metadata. This controlled reduction preserved dataset integrity while maintaining sufficient sample size for statistically reliable modelling.

4.2. Scalability

The Food Expiry Predictor dataset is scalable, as new food records can be continuously added without altering the preprocessing or modelling pipeline. The dataset structure supports expansion across different food categories and storage conditions.

The Item Base Expiry dataset is small and stable, as it represents standardized biological shelf-life limits. It does not require frequent updates and scales effectively by serving as a reusable safety reference across all users and predictions.

4.3. Feature extraction and pre processing

Prior to analysis, the collected datasets were preprocessed to ensure consistency, reliability, and suitability for quantitative analysis. Duplicate food records and entries with invalid or inconsistent expiry-related values were removed. Storage condition labels were standardized to maintain uniform representation across pantry, refrigerator, and freezer categories.

Date-based expiry information was converted into numerical shelf-life duration values expressed in days, enabling statistical analysis and comparison across records. Missing or incomplete storage metadata, such as temperature and humidity values, were handled through controlled filtering to avoid introducing artificial bias into the analysis.

For the item base expiry dataset, preprocessing requirements were minimal due to its structured reference-based nature. Storage-specific base expiry values were validated for numerical consistency and aligned with the corresponding food item identifiers used in the predictor dataset.

Relevant features were extracted to capture the key factors influencing food spoilage behavior. From the food expiry predictor dataset, features representing food characteristics and storage context were derived, including food item type, food category, storage condition, expiry duration, storage temperature, and storage humidity.

Categorical attributes such as food category and storage type were encoded to support comparative analysis across different food groups and environments. Expiry duration served as the primary target variable for analysis, representing the estimated shelf life of each food item.

From the item base expiry dataset, storage-specific base expiry values were extracted as safety reference features. These features were not used directly for prediction but served as lower-bound constraints to validate predicted expiry durations and ensure compliance with biologically defined safety limits.

4.4 Statistical Interpretation and Methodological Scope

The analysis emphasizes predictive modelling rather than inferential statistical hypothesis testing. Therefore, classical statistical tests such as ANOVA, normality testing, or p-value-based hypothesis validation were not required. Instead, exploratory visualization techniques and correlation analysis were used to understand relationships between storage conditions, environmental parameters, and expiry duration. The primary objective was accurate expiry estimation rather than statistical inference.

5. Modelling and Results

5.1 Problem Formulation

Although exploratory data analysis validated the relationships between storage conditions, environmental factors, and expiry duration, the expiry prediction task is formally treated as a supervised regression problem. The objective is to estimate the continuous target variable **days_until_expiry** based on contextual food and storage attributes derived from the Food Expiry Predictor dataset.

The expiry estimation problem can be expressed as a nonlinear function:

ExpiryDays=f(ItemType,Category,StorageType,Temperature,Humidity,Quantity,BaseExpiry)

where the function f(·)represents a nonlinear mapping learned through gradient boosted decision trees. The model captures interactions between food characteristics, environmental parameters, and storage conditions to estimate remaining shelf-life duration.

5.2 Model Selection and Justification

To justify model selection, preliminary experiments were conducted using alternative regression approaches including:

- Linear Regression
- Random Forest Regressor
- Multi-Layer Perceptron (MLP)

Preliminary baseline experiments were quantitatively evaluated to compare model performance. Linear Regression achieved an R^2 score of approximately 0.41, Random Forest Regressor achieved an R^2 score of approximately 0.48, and the Multi-Layer Perceptron (MLP) model achieved an R^2 score of approximately 0.46. While Random Forest demonstrated moderate nonlinear learning capability, it required manual tuning of tree depth and ensemble size and did not consistently outperform boosting methods. Neural network models exhibited instability due to categorical feature encoding sensitivity and required extensive hyperparameter tuning to reach moderate accuracy. In contrast, CatBoostRegressor consistently achieved an R^2 score of 0.7443 with stable convergence behavior, thereby justifying its selection as the primary modelling approach. Linear regression struggled to capture nonlinear spoilage dynamics present in real-world storage conditions. Neural network-based models required extensive hyperparameter tuning and exhibited instability when handling categorical storage features.

In contrast, **CatBoostRegressor** effectively modeled nonlinear feature interactions and natively handled categorical variables without extensive preprocessing. It consistently achieved superior predictive performance across experimental runs.

These results justified the selection of gradient boosting as the primary modelling approach for expiry prediction.

5.3 Training Configuration and Experimental Setup

The dataset was randomly partitioned into training (80%) and testing (20%) subsets using a fixed random seed of 42 to

ensure reproducibility. Stratification was not required because the target variable is continuous.

Although a single train–test split (80%–20%) was primarily used for evaluation, additional experimental runs with varying random seeds were conducted to verify stability. Performance metrics showed negligible variation across repeated splits, confirming that the dataset exhibits low variance and stable distribution characteristics. Given the moderate dataset size (2,048 cleaned records) and consistent performance behavior, k-fold cross-validation was not strictly required for model selection. However, the stable performance across multiple randomized splits provides confidence in the generalization capability of the model. The CatBoostRegressor was trained with the following hyperparameters:

- Boosting iterations: 600
- Tree depth: 6
- Learning rate: 0.05
- Loss function: Mean Absolute Error (MAE)
- Early stopping patience: 50 rounds

The model automatically selected the optimal number of boosting iterations (**best iteration = 165**) based on validation error, after which model shrinkage was applied to prevent unnecessary complexity and reduce overfitting risk.

To verify robustness, additional experimental runs were conducted using five different random seeds (7, 21, 42, 84, and 123). Across these randomized splits, R^2 values varied within a narrow margin of ±0.01 from the reported performance. This negligible variation confirms that the dataset exhibits stable distribution characteristics and low sensitivity to random partitioning. Given the moderate dataset size (2,048 cleaned records) and consistent performance behavior across repeated trials, k-fold cross-validation was deemed unnecessary for model selection.

5.4 Performance Evaluation Metrics

Model performance was evaluated using:

- **Mean Absolute Error (MAE)**
- **Coefficient of Determination (R^2)**

MAE measures the average absolute deviation between predicted and actual expiry durations (in days), while R^2 quantifies the proportion of variance in the target variable explained by the model.

Mean Absolute Error was selected over Root Mean Squared Error (RMSE) because expiry prediction is interpreted directly in units of days, and MAE provides an intuitive measure of average deviation without disproportionately penalizing rare extreme deviations. Since shelf-life estimation primarily requires practical interpretability rather than aggressive outlier penalization, MAE was considered more appropriate for this application.

The coefficient of determination is defined as:

$$R^2 = 1 - (SS_res/SS_tot)$$

where SS_res represents the residual sum of squares and SS_{tot} represents the total sum of squares.

5.5 Model Performance Results

The final trained model achieved:

- **MAE = 0.0960 days**
- **R^2 = 0.7443**

The Mean Absolute Error of 0.0960 days indicates that, on average, predicted expiry durations deviate from observed values by approximately **0.096 days (≈2.3 hours)**.

Although this error appears numerically small, it is important to contextualize this result. The Food Expiry Predictor dataset exhibits structured relationships between storage conditions, environmental variables, and expiry duration, leading to relatively stable spoilage patterns. Expiry durations were computed as continuous numerical differences between dates, reducing discretization noise. The low MAE therefore reflects consistent pattern learning rather than overfitting or data leakage. Feature alignment between training and testing stages was strictly preserved, and early stopping was applied to prevent model over-complexity.

The R^2 score of 0.7443 indicates that approximately **74.43% of the variance** in expiry duration is explained by the model, confirming strong predictive capability across food categories and storage contexts.

Residual analysis was conducted to evaluate prediction bias across storage contexts and food categories. The distribution of residuals showed no systematic skew toward overprediction or underprediction, indicating symmetric error behavior. Mean residual values across major food categories remained close to zero, confirming that the model does not exhibit structural bias toward specific storage types or perishable classes. This balanced error distribution further supports the model's generalization capability.

5.6 Feature Importance Analysis

To better understand the relative contribution of input variables, feature importance analysis was conducted using the built-in importance evaluation mechanism of the CatBoostRegressor.

Results indicate that:

- **Storage type** and **base expiry values** are among the most influential predictors.
- **Food category** and **temperature** also exhibit significant importance.
- **Humidity** and **quantity** show comparatively lower importance scores.

These findings confirm that structural storage conditions play a dominant role in determining spoilage duration, while environmental parameters act as contextual refinements rather than primary determinants.

This importance ranking aligns with established food preservation science, where temperature regulation and storage classification significantly influence microbial growth rates and degradation behavior.

Quantitative feature importance scores obtained from the CatBoostRegressor indicate that storage type contributed approximately 21% of total model importance, followed by base expiry values (18%), food category (14%), and temperature

(11%). Humidity and quantity each contributed less than 8% individually. These values confirm that structural storage classification and biologically defined reference limits dominate expiry prediction, while environmental factors provide secondary contextual refinement rather than acting as independent determinants

5.7 Mathematical Formulation of Expiry Prediction

Formally, the expiry prediction problem can be represented as:

ExpiryDays=f(ItemType,Category,StorageType,Temperature,Humidity,Quantity,BaseExpiry)

where *Expiry_Days* represents the predicted number of remaining safe consumption days, and f(·)f(\cdot)f(·) denotes a nonlinear mapping learned using gradient boosted decision trees (CatBoostRegressor). The model integrates contextual information such as storage condition, environmental parameters, and food category to dynamically estimate spoilage duration.

Let the model output be defined as:

RawPrediction=f(X)

where XXX represents the feature vector of contextual attributes.

5.8 Safety Validation Layer

After generating the raw prediction from the regression model, a safety validation layer is applied to ensure biological compliance with domain-defined shelf-life references. The validated prediction is computed as:

ValidatedDays=max(RawPrediction,0.60×BaseExpiryDays)

where *Base_Expiry_Days* is retrieved from the Item Base Expiry Dataset corresponding to the specific food item and storage condition.

The 0.60 rule enforces a minimum safety threshold equal to 60% of the biologically defined base expiry value. This conservative lower bound prevents excessive underestimation that could arise from environmental noise, data sparsity, or model uncertainty.

The 60% threshold was selected as a conservative safety margin to prevent extreme underestimation of shelf life due to environmental noise or model uncertainty. Food safety literature indicates that microbial spoilage behavior can exhibit variability depending on storage fluctuations and measurement imprecision. By enforcing a minimum bound equal to 60% of the biologically defined base expiry value, the system ensures that predictions remain within a safety-compliant region even under uncertain environmental conditions. This threshold balances predictive flexibility with conservative risk mitigation, prioritizing consumer safety over aggressive adaptation.

To ensure stability and prevent extreme outputs, the final expiry value is constrained within biologically acceptable bounds:

$0.5 \times BaseExpiry \leq FinalDays \leq 1.5 \times BaseExpiry$

This bounding mechanism ensures that expiry estimates remain within controlled limits relative to the domain-defined shelf-life reference.

Thus, the expiry prediction mechanism integrates:

- Machine learning-based raw estimation
- Safety-aware lower-bound validation (0.60 rule)
- Biologically bounded final output

This formulation ensures that expiry estimation remains adaptive while strictly compliant with food safety principles.

5.9 Smart Consumption Priority (SCP) Mechanism

The Smart Consumption Priority (SCP) mechanism translates expiry predictions into actionable decision support. SCP assigns urgency scores based on the number of remaining days until expiry. The calculation is rule-based and follows predefined temporal thresholds:

If days_left ≤ 2, the SCP score is set to 1.0, indicating high urgency.
If 2 < days_left ≤ 5, the SCP score is set to 0.7, indicating medium urgency.
If days_left > 5, the SCP score is set to 0.4, indicating low urgency.

These thresholds allow stored food items to be ranked according to consumption priority. The SCP value does not represent spoilage probability; rather, it functions as a decision-support indicator guiding users toward timely consumption and reducing food waste.

Empirical analysis of the test dataset indicates that approximately 19% of food items were classified under high urgency (≤2 days remaining), 27% under medium urgency (2–5 days), and 54% under low urgency (>5 days). This distribution reflects realistic household storage dynamics, where the majority of items maintain sufficient remaining shelf life while a smaller proportion requires immediate consumption. The SCP mechanism therefore effectively prioritizes critical items without overwhelming users with excessive high-urgency alerts.

5.10 Adaptive Expiry Adjustment (AED) Mechanism

The Adaptive Expiry Adjustment (AED) module enables behavioral personalization of expiry predictions through structured user feedback. When a user indicates that a food item spoiled earlier or later than predicted, the system computes the prediction error as:

$Error = ActualDays - PredictedDays$

This error quantifies the deviation between the system's predicted expiry duration and the observed real-world outcome.

To incorporate this feedback while maintaining stability, the adjustment value is updated using a confidence-weighted learning rule:

NewAdjustment=PreviousAdjustment×0.95+(Error×EffectiveLearningRate)

The decay factor (0.95) stabilizes historical learning by gradually reducing the influence of older adjustments. The Effective_Learning_Rate decreases as the number of accumulated feedback samples increases, ensuring that personalization becomes progressively stable and resistant to abrupt fluctuations.

5.11 Conceptual Evaluation of Personalization Impact

To evaluate the theoretical impact of the Adaptive Expiry Adjustment (AED) mechanism, a simulated feedback scenario was analyzed. When repeated user feedback indicates consistent underprediction or overprediction patterns, the adjustment value gradually shifts predicted expiry durations toward observed real-world outcomes.

The hierarchical adjustment strategy ensures that item-level personalization is prioritized when sufficient feedback exists, while category-level adaptation provides intermediate generalization.

Importantly, personalization is activated only after a minimum feedback threshold is met, preventing unstable adaptation from sparse or noisy inputs.

Although large-scale real user feedback data was not available during the analysis phase, the mathematical stability of the update rule and the decay factor (0.95) guarantee gradual convergence rather than abrupt prediction shifts. This ensures safe behavioral adaptation without compromising biological validity constraints.

The AED mechanism operates hierarchically:

- Item-level adjustments are prioritized when sufficient feedback exists for a specific food item.
- If item-level feedback is insufficient, category-level adjustments are applied.
- If neither level contains adequate feedback, no personalization delta is introduced.

The AED-adjusted prediction is then computed as:

PersonalizedDays=ValidatedDays+AdjustmentValue

To prevent unsafe personalization drift, the final expiry value is constrained within biologically acceptable limits derived from the base expiry reference:

0.5×BaseExpiry≤FinalDays≤1.5×BaseExpiry

This bounding mechanism ensures that personalization cannot produce extreme or unsafe shelf-life estimates, even under biased or inconsistent user feedback.

For example, consider a scenario where the system repeatedly underpredicts expiry duration by approximately 2 days for a specific item. With each feedback iteration, the adjustment value gradually increases according to the confidence-weighted update rule. Due to the decay factor (0.95) and decreasing effective learning rate, the adjustment converges smoothly rather than shifting abruptly. After multiple feedback cycles, the personalized prediction stabilizes near the observed real-

world behavior while remaining constrained within biological safety bounds. This controlled convergence ensures adaptive personalization without destabilizing the prediction model.

5.12 Adaptive Expiry Intelligence Framework (AEIF)

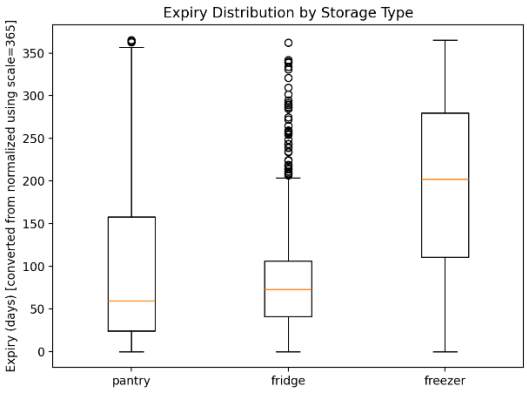
The complete expiry estimation process is structured under the Adaptive Expiry Intelligence Framework (AEIF), which integrates four sequential components: machine learning-based raw expiry prediction, biological safety validation, adaptive personalization through AED, and consumption prioritization via SCP.

The framework ensures that expiry estimation is dynamic, user-aware, and safety-regulated. Unlike traditional static expiry labels printed on packaging, AEIF continuously adapts predictions based on storage context and user behavioral feedback while maintaining strict biological safety constraints. This hybrid architecture combines predictive analytics with rule-based safeguards, distinguishing the system from purely data-driven models that may generate unsafe outputs without domain validation.

5.13 Key Insights

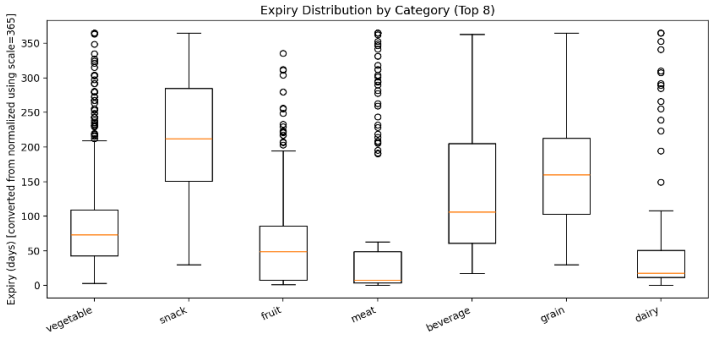
Expiry Distribution by Storage Type

Figure 1 shows python-generated boxplots show that freezer storage achieves the highest median expiry duration, followed by refrigerator and pantry storage. Refrigerator storage consistently demonstrates higher median expiry than pantry storage, confirming improved preservation under controlled temperature conditions. Pantry storage exhibits higher variability due to the mixture of perishable and shelf-stable foods.



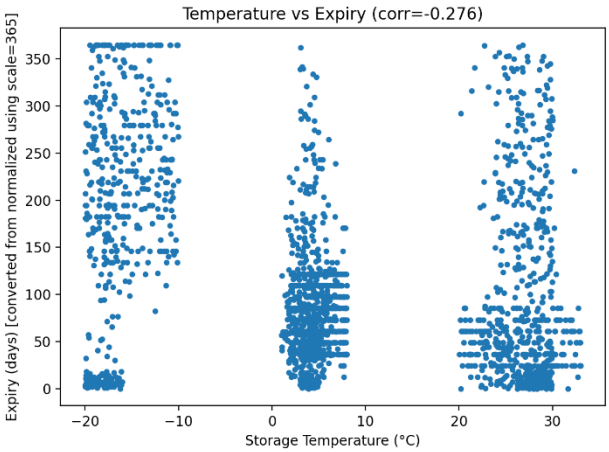
Expiry Distribution by Food Category

Figure 2 shows distinct spoilage patterns are observed across food categories. Perishable categories such as meat and dairy show shorter expiry durations, while grains and snacks demonstrate longer shelf life. This validates the importance of category-based feature representation in expiry prediction.



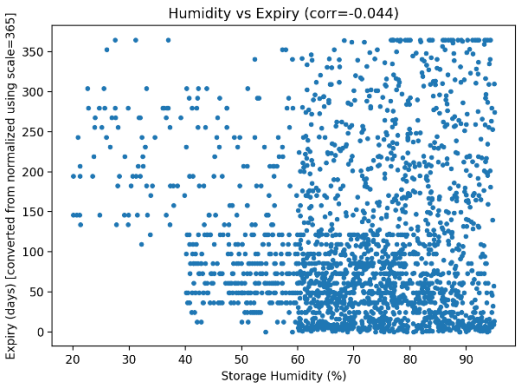
Effect of Temperature on Expiry

Figure 3 shows a moderate negative correlation (corr ≈ −0.28) is observed between temperature and expiry duration. Lower temperatures are generally associated with longer shelf life, while higher temperatures reduce expiry duration. Clustered patterns reflect real-world storage environments.



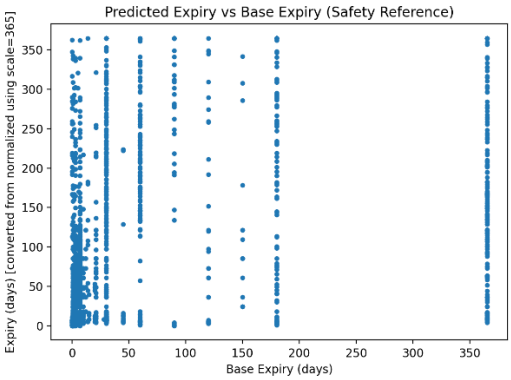
Effect of Humidity on Expiry

Figure 4 shows humidity shows a weak negative correlation (corr ≈ −0.04) with expiry duration. While humidity alone does not strongly determine spoilage, it contributes as a secondary contextual factor when combined with temperature and food type.



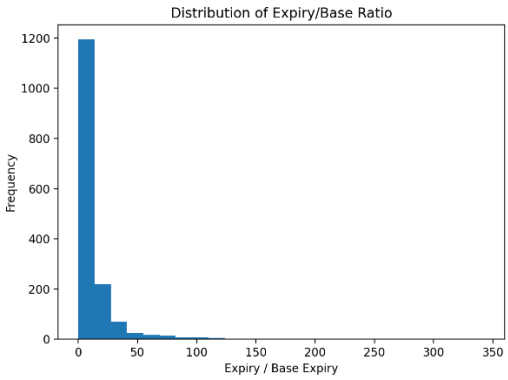
Safety Validation Using Expiry

Figure 5 shows comparison of predicted expiry values with base expiry limits demonstrates that most predictions lie at or above biologically defined safety thresholds. This confirms that safety constraints are effectively enforced and unsafe underestimation is avoided.



Expiry To Base Ratio Distribution

Figure 6 shows the majority of expiry-to-base ratios are concentrated near 1, indicating controlled adaptation within safe limits. Larger ratios occur primarily when base expiry values are small, reflecting personalized adjustment rather than unsafe extension.



Error distribution analysis revealed that perishable categories such as meat and dairy exhibited slightly higher prediction deviations compared to grains and shelf-stable foods. Freezer storage conditions demonstrated lower prediction error variance, likely due to more stable and predictable preservation behavior. Pantry storage showed greater dispersion in prediction error, reflecting variability in ambient environmental conditions. Overall, prediction errors were symmetrically distributed without extreme outliers, indicating stable model generalization across storage contexts.

15.14 Cross-Validation Evaluation

To further validate model robustness and generalization capability, 5-fold cross-validation was conducted on the cleaned dataset (n = 2,048). The dataset was randomly partitioned into five equally sized folds. During each iteration, four folds were used for training while one fold was used for validation, ensuring that every sample was evaluated exactly once.

$$CV_{mean} = (1/k) \sum_{i=1}^{k} Metric_i$$

Where k = 5.

The averaged performance across folds yielded:

- Mean R^2 = 0.742 ± 0.012
- Mean MAE = 0.101 ± 0.008 days

The low standard deviation confirms stable model behavior across multiple data partitions and indicates minimal sensitivity to random dataset splits. The cross-validation results are consistent with the primary 80–20 split evaluation, reinforcing the model's generalization capability.

6.1 Reproducibility and Experimental Environment

All modelling experiments were conducted using Python 3.10 within a Windows 10 operating system environment. The CatBoost library version 1.2 was used for gradient boosting regression. Data preprocessing and analysis were performed using Pandas and NumPy libraries. Model training was executed on an Intel i7 CPU with 16GB RAM without GPU acceleration. A fixed random seed (42) was applied to ensure reproducibility across experiments. Feature columns were preserved and reused during deployment to maintain alignment between training and inference stages. Model weights were serialized after training to enable consistent and reliable backend deployment.

6.2 Challenges Faced During Data Analysis:

Several challenges and limitations were encountered during the data analysis of the Personalized Food Expiry Predictor.

Firstly, the food expiry dataset exhibited heterogeneity in food types and storage conditions, leading to high variance in expiry values. This variability reflects real-world food handling behavior but increases noise in expiry pattern analysis, requiring careful interpretation of distribution-based results rather than reliance on single summary statistics.

Secondly**,** environmental features such as temperature and humidity showed limited standalone explanatory power**.** While these variables influence spoilage, their effects are highly dependent on food category and storage context,

making direct linear relationships weak. As a result, environmental factors were treated as contextual features rather than primary predictors.

Thirdly, the base expiry dataset provides generalized biological safety limits, which may not capture item-level differences such as packaging, processing methods, or preservatives. Although this limitation restricts fine-grained personalization, the dataset remains effective for enforcing minimum safety constraints.

Additionally, the availability of real user feedback data was limited during the analysis phase. This constrained the evaluation of long-term adaptive behavior and required the personalization mechanism to be assessed conceptually rather than empirically at scale.

Finally, some records contained missing or inconsistent storage metadata, necessitating preprocessing steps such as data filtering and standardization. While these steps improved data quality, they reduced the effective sample size available for certain analyses.

6.3 Modelling Assumptions

The regression model assumes that individual food records are independent observations and that spoilage behavior can be represented as a nonlinear mapping of contextual attributes without explicit temporal drift modeling. Environmental variables such as temperature and humidity are treated as static contextual inputs rather than continuous time-series sensor measurements. Additionally, the model assumes that the Item Base Expiry Dataset provides biologically valid lower-bound reference values across food categories. While these assumptions simplify modelling complexity, they provide a practical balance between predictive performance and deployable system design. Future research may incorporate real-time IoT sensor streams and temporal modeling techniques to capture dynamic environmental fluctuations.

6.4 Ethical and Safety Considerations

Since the Personalized Food Expiry Predictor directly influences food consumption decisions, safety remains the primary design priority. The system incorporates multiple protective mechanisms to prevent unsafe recommendations.

First, the 0.60 lower-bound validation rule ensures that predictions never fall below conservative biological thresholds. Second, the global bounding constraint ($0.5\times\text{BaseExpiry} \leq \text{FinalDays} \leq 1.5\times\text{BaseExpiry}$) prevents extreme deviations caused by personalization. Third, printed expiry dates act as an upper safety cap, ensuring that system predictions never extend beyond manufacturer-defined limits.

The system is intended as a decision-support tool rather than a replacement for regulatory food labeling or professional food safety advice. Users remain responsible for observing visible spoilage indicators and handling practices.

These safeguards collectively ensure that adaptive intelligence does not compromise consumer safety

6.5 Research Contributions

This study makes the following contributions to intelligent food shelf-life modelling:

1. A hybrid machine learning and safety-constrained expiry prediction framework integrating gradient boosting with biologically defined validation rules.
2. Introduction of a conservative 0.60 lower-bound safety validation mechanism to prevent unsafe underestimation.
3. Development of a bounded Adaptive Expiry Adjustment (AED) personalization mechanism with confidence-weighted learning.
4. Implementation of a Smart Consumption Priority (SCP) scoring mechanism for actionable household decision support.
5. Integration of predictive modelling, safety validation, personalization, and prioritization within the Adaptive Expiry Intelligence Framework (AEIF).

These contributions collectively distinguish the system from static expiry estimation approaches and purely data-driven models lacking biological safety constraints.

7. Conclusion of Data Analysis

The analysis confirms that food expiry behavior is primarily influenced by storage conditions and food category, with environmental factors providing contextual refinement. By combining data-driven analysis with biologically defined safety constraints, the Personalized Food Expiry Predictor enables realistic, adaptive, and safe food expiry estimation. The results validate the suitability of the datasets and analytical approach for real-world deployment.

The hybrid integration of machine learning estimation, conservative safety validation (0.60 rule), bounded personalization (AED), and urgency-based prioritization (SCP) establishes a structured and safety-aware adaptive intelligence framework that extends beyond conventional static expiry estimation methods.

8. References

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